ClipSAT : Multivariable Calculus Partials, Multiple Integrals & Vector Calcu

Unit 3.01: Multivariable Optimization, Hessians & Lagrange Multipliers

Name: _____

Q1. Consider the function z = fx,y = x^2 + y^2 . Let x = 2t and y = 3t . Use the Multivariable Chain Rule to find frac{dz}{dt} when t=1 .

[ ] A: 10
[ ] B: 13
[ ] C: 26
[ ] D: 52

Q2. Evaluate the double integral of the function fx,y = 2xy over the rectangular region R = [0, 2] times [0, 3] using Fubini's Theorem. Set up the iterated integral by integrating with respect to y first, then x .

[ ] A: 12
[ ] B: 18
[ ] C: 6
[ ] D: 24

Q3. For fx,y=7 x^{2} + 4 y^{2} , find f_y-3,1 .

[ ] A: -24
[ ] B: 4
[ ] C: 8
[ ] D: -42

Q4. Consider the function fx,y = x^2 + y^2 . Find the value of f_{xx}2,3 .

[ ] A: 2
[ ] B: 4
[ ] C: 6
[ ] D: 0

Q5. Let the linear scalar potential be hu, y = 15u + 8y - 16 . Compute the squared gradient norm |nabla h|^2 .

[ ] A: 289
[ ] B: 293
[ ] C: 281
[ ] D: 304

Q6. Consider the surface S which is the portion of the plane z = 2x + 3y + 1 lying above the rectangular region R = [0, 1] times [0, 2] in the xy -plane. Calculate the surface area of S .

[ ] A: sqrt{14}
[ ] B: 2 sqrt{14}
[ ] C: 3 sqrt{14}
[ ] D: 4 sqrt{14}

Q7. Consider the vector field mathbf{u}s, y, z = langle 22s,, 7y,, 7z rangle on mathbb{R}^3 . Compute the scalar divergence nabla cdot mathbf{u} .

[ ] A: 58
[ ] B: 40
[ ] C: 29
[ ] D: 36

Q8. Consider the vector field mathbf{F}x,y = langle 3x, 2y rangle . Evaluate the vector field at the point 2, 4 .

[ ] A: langle 6, 8 rangle

[ ] B: langle 8, 6 rangle

[ ] C: langle 5, 6 rangle

[ ] D: langle 6, 5 rangle